Set Top Box Multiplayer Framework

Team: Plugged-In
Marshall Gaucher, David Jarman, Kevin Worner
Sponsored by: Innovative Systems, L.L.C

Client

- Innovative Systems
  - Provides integrated solutions for the independent telecommunications industry.
- APMax
  - Application Server providing services such as:
    - IPTV Video Solutions
    - Unified Messaging
    - Single Number Service
- STB
  - Set-top-box providing television and integrated telephony applications

Project

- Create a server to host STB multiplayer game plugins
- Design a template game UI for STBs in an APMax environment
- Build a virtual APMax to simulate future communication protocols
- Convert games to proof-of-concept multiplayer games.

Sprint Review

- Sprint #1
  - Initial design of STBMF system
  - Researched current Innovative System network communications
- Sprint #2
  - Designed prototypes of STBMF System (vSTB, vAPMAX, Game Server)
  - Designed communication definitions for STBMF
- Sprint #3
  - Implemented prototypes to test designed communication definitions
  - Created sand box environment for development

System Design

- 3 Major Components:
  - vSTB
  - vAPMAX
  - Game Server
- Communication:
  - UDP and TCP
  - CommandLink protocol

Prototypes

- vSTB
  - Scalable module to handle game environment request/response from user
- vAPMAX
  - Message handler between vSTB and Game Server
- Game Server
  - Process requests from/responses to users on vSTB

vSTB

- New request response definitions
- Scalable
- Modular (vSTB Tester, vSTB Multiplayer)
- Platform independent
- Increased availability to Users

User Story

- vSTB must work on actual STB

Amino Live

Connectivity:
Dual gigabit Ethernet via RJ-45 shielded connector, 802.11n wireless, Bluetooth, sim or smart card

Outputs:
HDMI 1.4 with HDCP. USB 2.0 8 3.0

Memory:
2 GB DDR 3 RAM

HDD:
250GB standard. Other sizes on request.

Codecs:
MPEG-2 MP@HL, MPEG-4 pt10 AVC/H.264 HP@L4
Multi-plane dual HD (1080p) decode and display of video (MPEG 2, VCI and H264) and other low bit rate codec support

CPU:
Intel® Dual Core Atom™ based with hyper threading

Graphics:
Advanced graphics core with OpenGL2.0 and 3D shader support

Amino A140

Connectivity:
Ethernet 10/100 BaseT via RJ-45 shielded connector

Outputs:
HDMI 1.3a. (excl. Deep colour and DTS audio) with HDCP.S/PDIF (optical). USB2.0. 10-way Mini-DIN for Composite video, Component (YPrPb),RGB, S-Video and analogue audio. RF Mod and loop through.

Codecs:
MPEG-2 MP@HL. MPEG-4 pt10 AVC/H.264 HP@L4

GRAPHICS RESOLUTIONS
HD graphics up to 1280x720

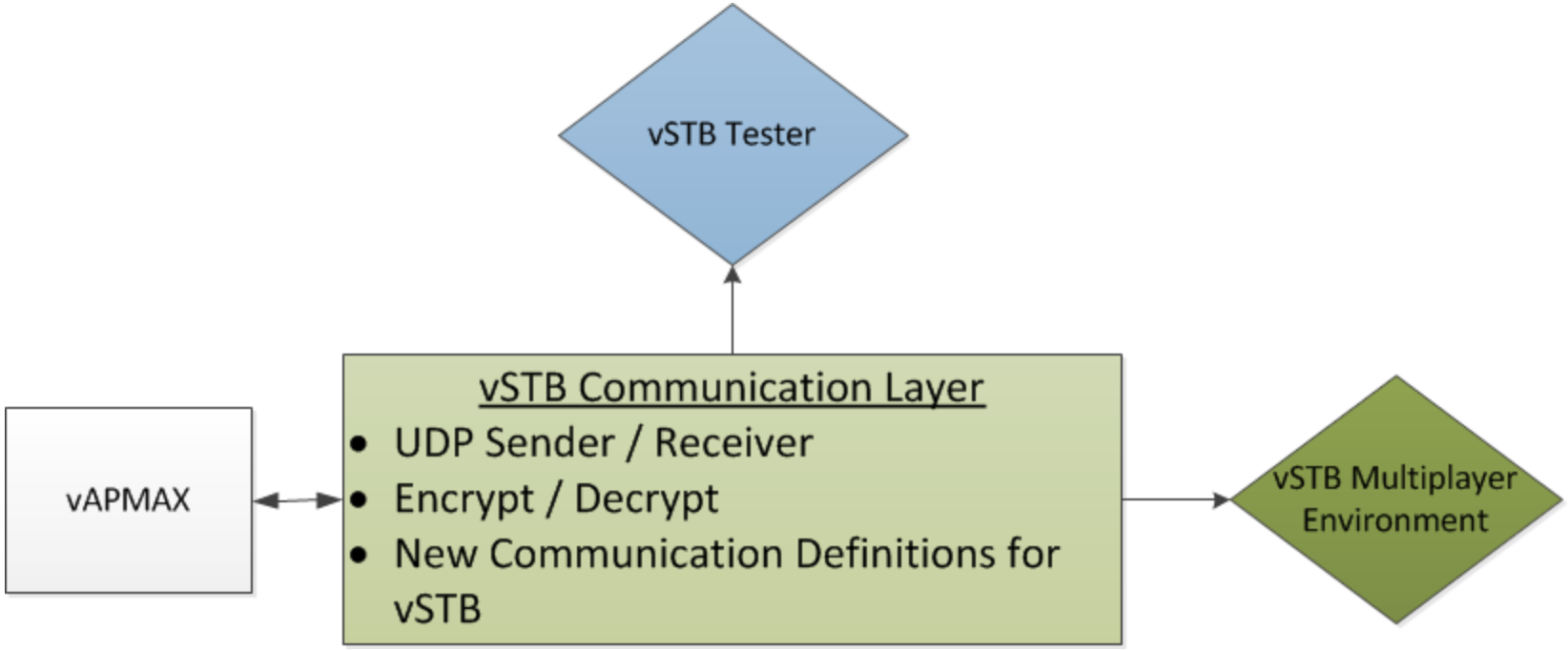
Memory:
128MB Flash, 256MB RAM

CPU:
800 Mhz Processor

User Story

- vSTB must be fully Unit Tested
- Unit Testing: QtTestLib
  - Lightweight library, around 6000 lines of code and 60 exported symbols.
  - Data-driven unit testing and GUI testing.
  - IDE Integration for Visual Studio and KDevelop.
  - Thread-safe error reporting.
  - Self-contained, requires only Qt Core for non-gui testing.
  - Rapid testing, no special runners required.
  - Rapid test creation, no special utilities or testcase-registration required.
  - Easily extendable test logging, outputs plain text or xml.
- QElapsedTimer
  - milliseconds, nanoseconds(best estimated by system)

vSTB Design



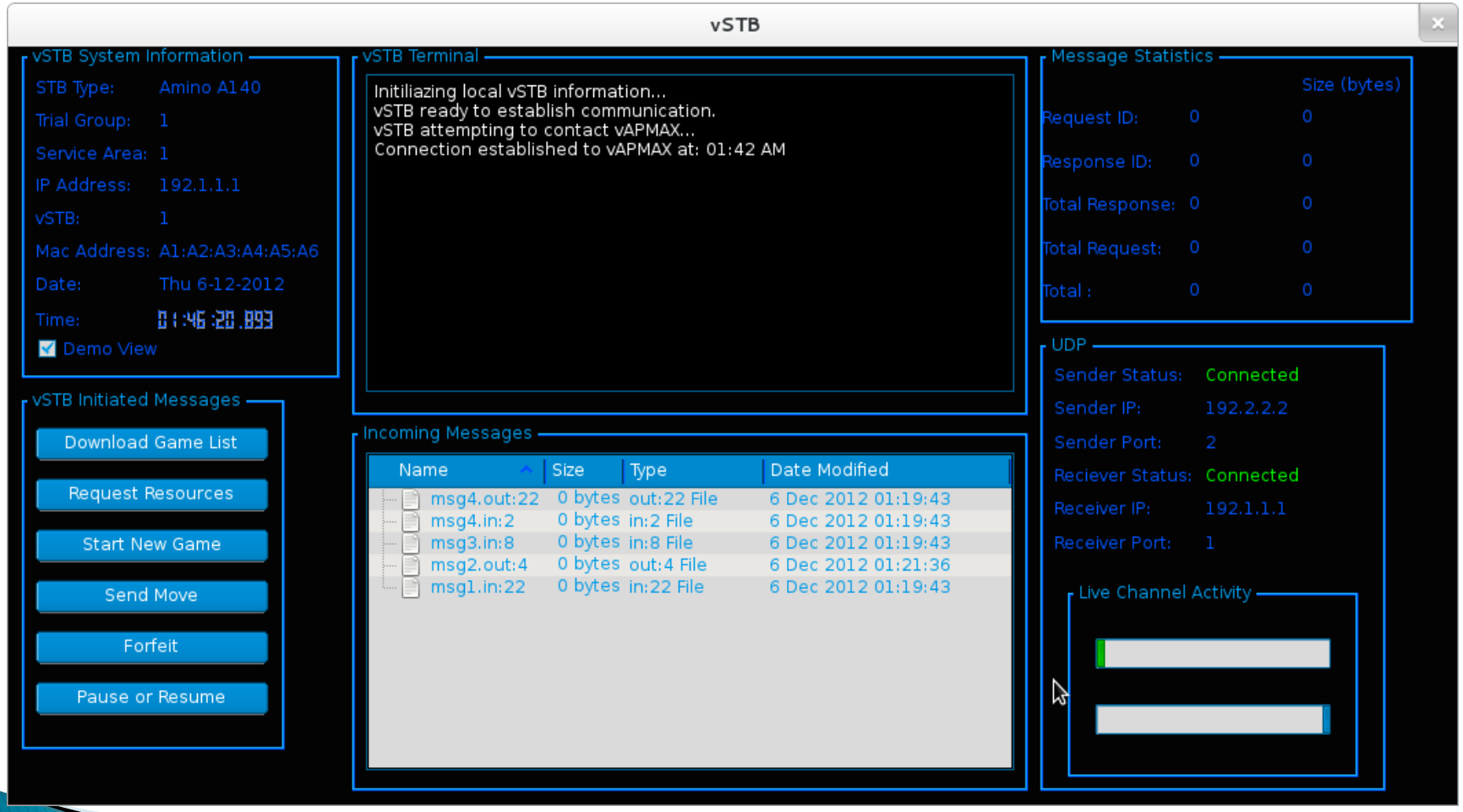
vSTB Tester

- Provides vAPMAX intuitive interface to test environment
- Provides Game Server a flexible tool to view deployed objects

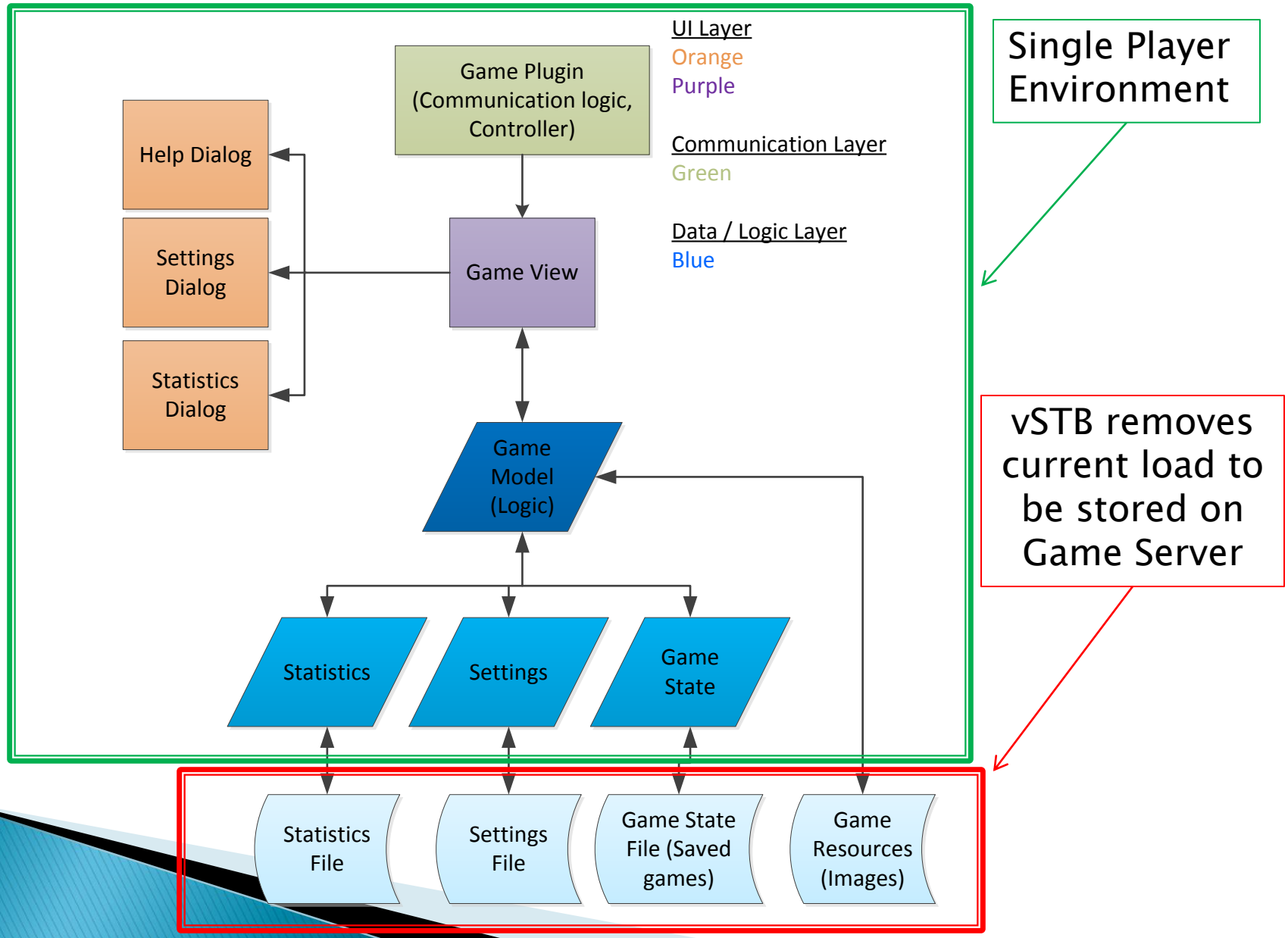
vSTB Multiplayer Environment

- Reduces current load on STB
- Framework provides foundation to add future features to meet demand of end user and technology

vSTB Tester


vSTB
vSTB System Information
STB Type: Amino A140
Trial Group: 1
Service Area: 1
IP Address: 192.1.1.1
vSTB: 1
Mac Address: A1:A2:A3:A4:A5:A6
Date: Thu 6-12-2012
Time: 01:46:20.893
Demo View
vSTB Initiated Messages
Download Game List
Request Resources
Start New Game
Send Move
Forfeit
Pause or Resume
vSTB Terminal
Initiliazing local vSTB information...
vSTB ready to establish communication.
vSTB attempting to contact vAPMAX...
Connection established to vAPMAX at: 01:42 AM
Incoming Messages
Name Size Type Date Modified
msg4.out:22 0 bytes out:22 File 6 Dec 2012 01:19:43
msg4.in:2 0 bytes in:2 File 6 Dec 2012 01:19:43
msg3.in:8 0 bytes in:8 File 6 Dec 2012 01:19:43
msg2.out:4 0 bytes out:4 File 6 Dec 2012 01:21:36
msg1.in:22 0 bytes in:22 File 6 Dec 2012 01:19:43
Message Statistics
Size (bytes)
Request ID: 0 0
Response ID: 0 0
Total Response: 0 0
Total Request: 0 0
Total : 0 0
UDP
Sender Status: Connected
Sender IP: 192.2.2.2
Sender Port: 2
Reciever Status: Connected
Receiver IP: 192.1.1.1
Receiver Port: 1
Live Channel Activity

Current STB Game Architecture



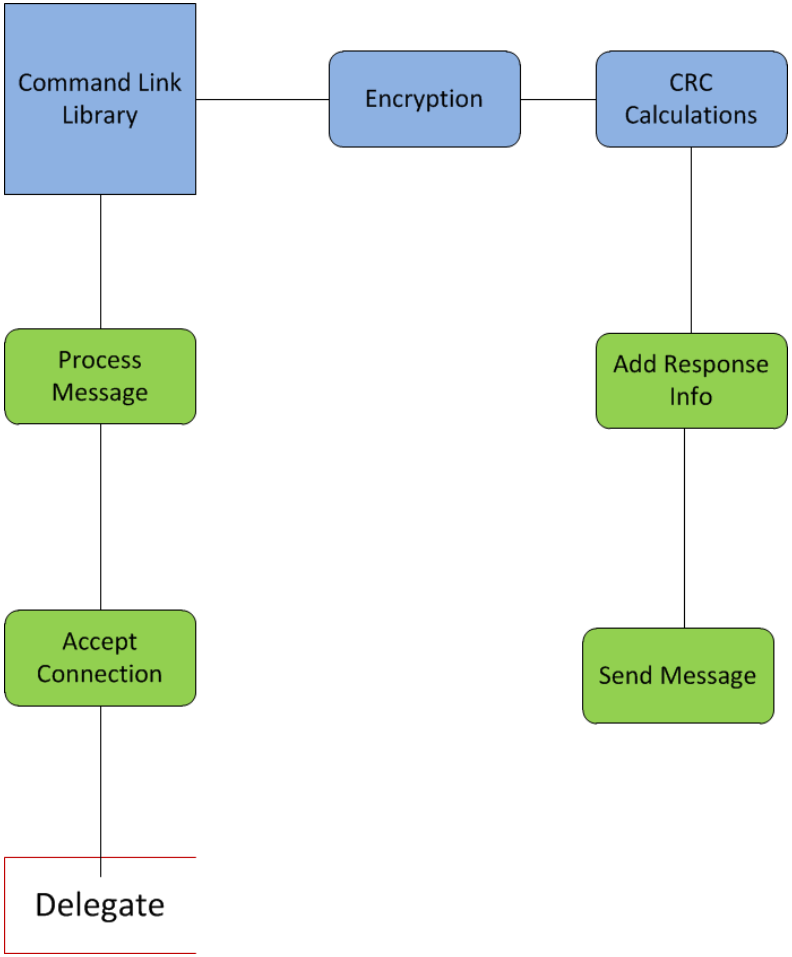
vSTB Direction

- Stabilize communication layer
- Redesign single player game architecture to exist with new live multiplayer environment

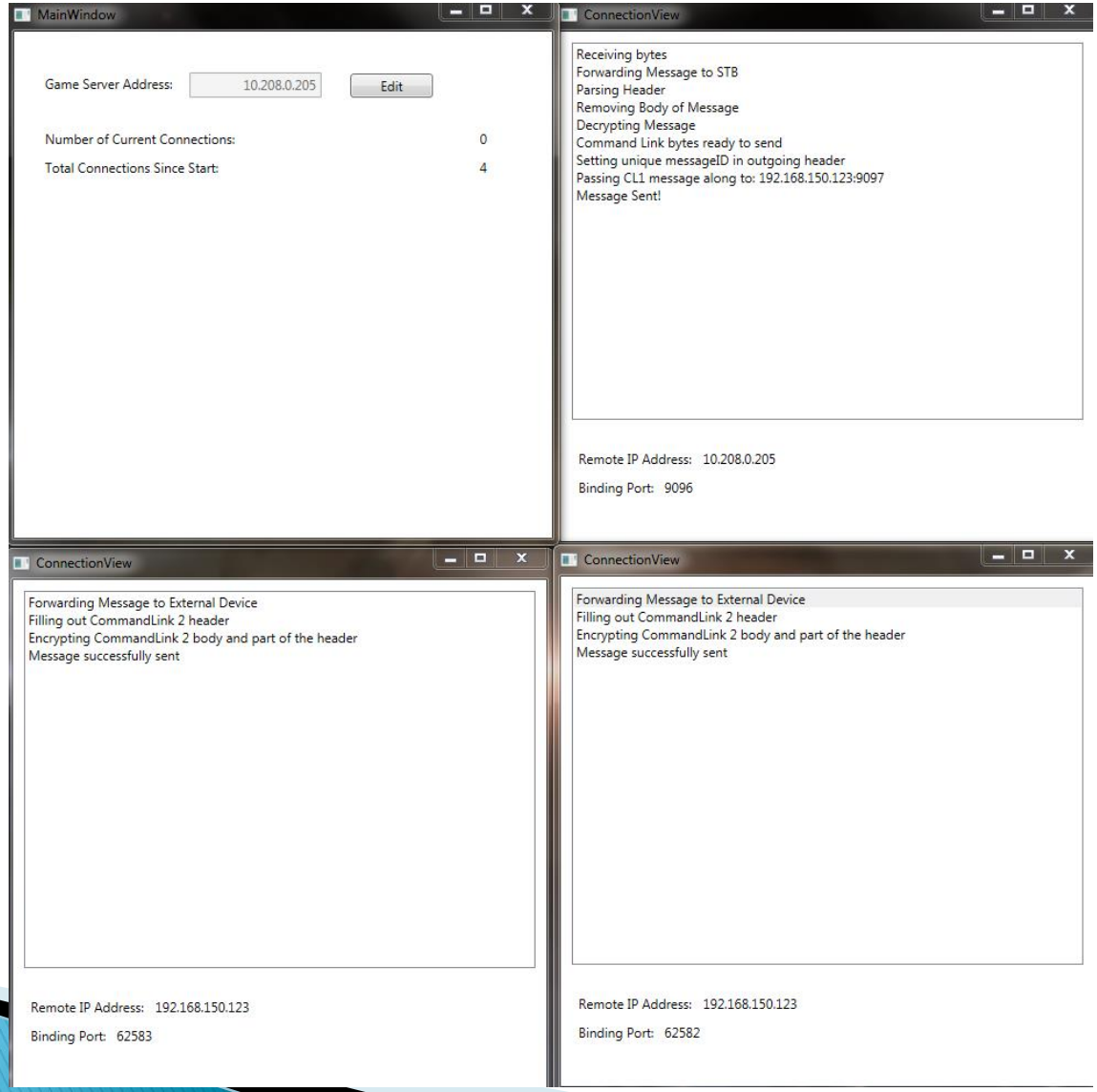
vAPMax

- Pass through messaging system
  - STB
  - Game Server
- Performs quick formatting
- Allows quicker development
- Can test new communication protocols

vAPMax Dataflow

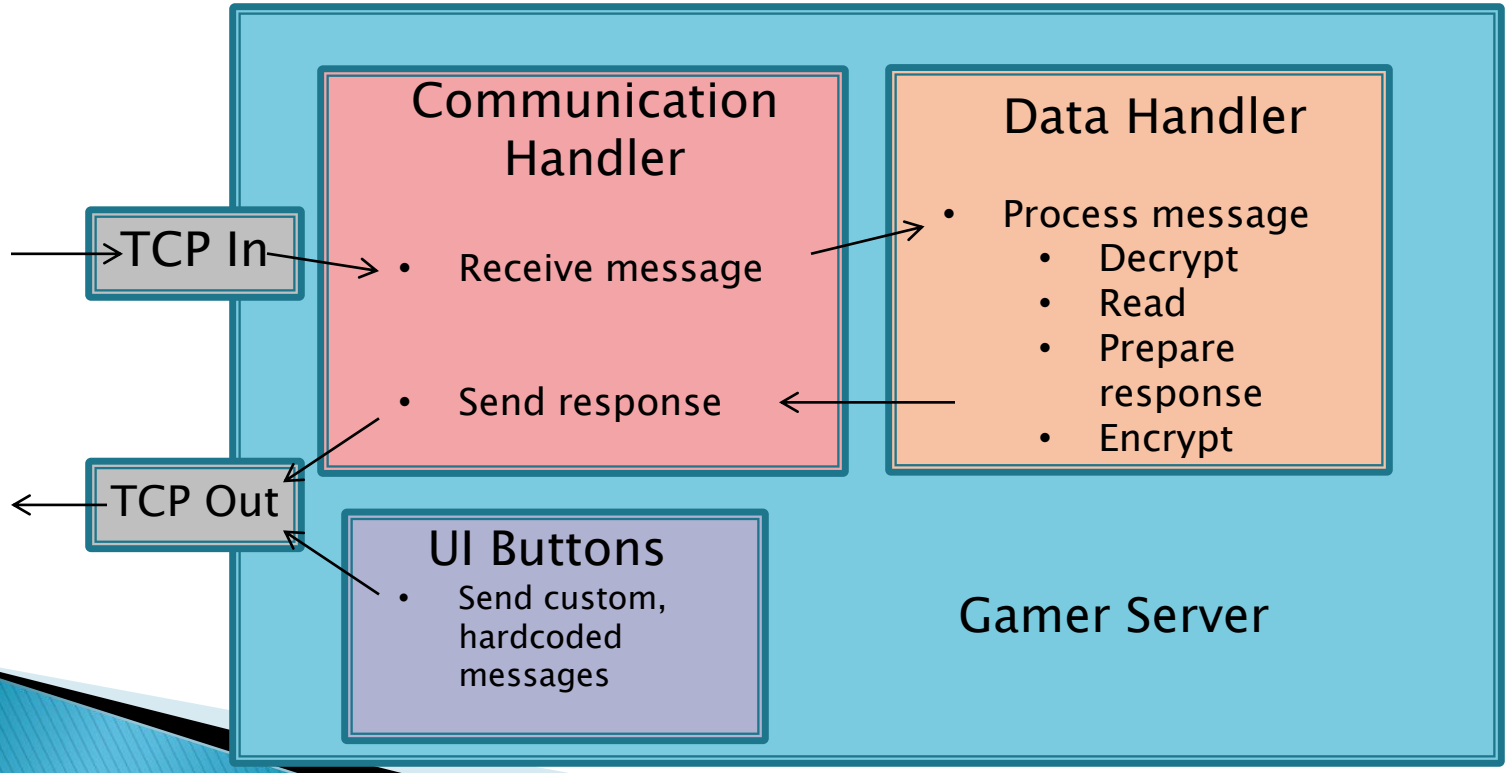


vAPMax Prototype

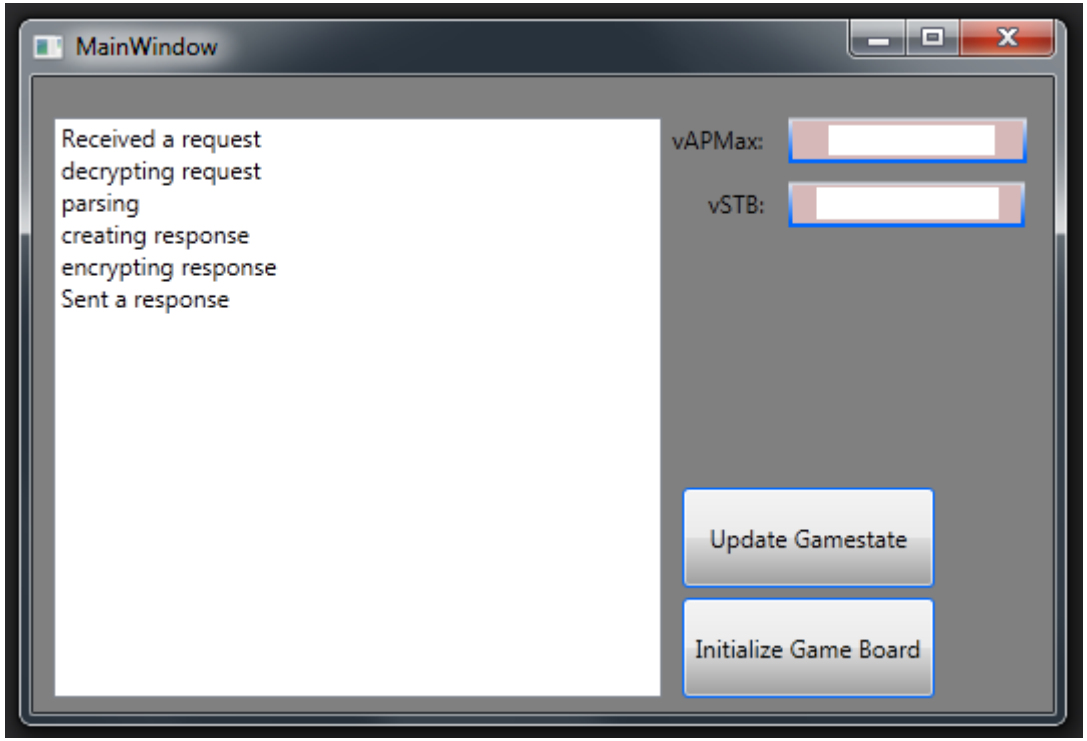


Game Server

- Will manage game resources/instances/stats
- Right now: Manages communication



Game Server Prototype



Product Backlog

- vSTB
  - Research Qt Unit Testing
  - Create Checkers game as a proof of concept
  - Prove out and implement QUiLoader
  - Research Dynamic UI
  - Refactor existing STB game view/model architecture to meet STBMF requirements
  - Create Memory game as a proof of concept
- vAPMAX
  - Integrate Logger into project and provide basic command line usage to view.
  - Unit Testing
  - Update Reactive Extension package to latest
- Game Server
  - Server: Plugin Interface
  - Server: Database
  - Server: UI – Stats, Import
  - Server: Future – server components, database
  - Server: Design: game interface – loading game, playing game
  - Server: Unit Test
  - Server: Build template GUI
  - Server: Research Server operating systems
  - Server: Research Security
  - Server: Convert to a service
  - Server: Research attaching a GUI to a service using C#

Questions